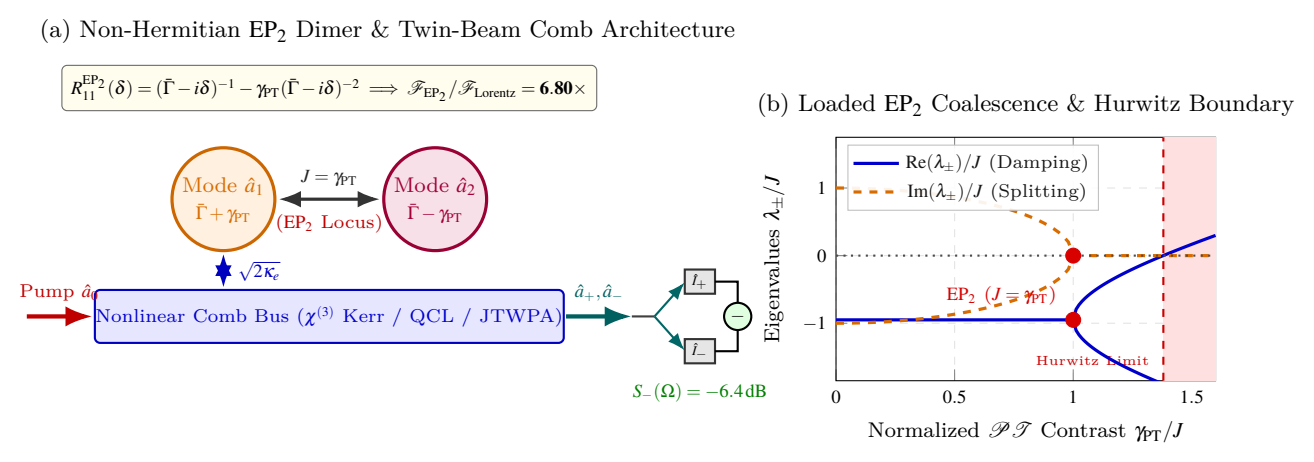
(a) Non-Hermitian EP$_2$ Dimer & Twin-Beam Comb Architecture
$R_{11}^{\mathrm{EP}_2}(\delta) = (\bar{\Gamma} - i\delta)^{-1} - \gamma_{\mathrm{PT}}(\bar{\Gamma} - i\delta)^{-2} \implies \mathscr{F}_{\mathrm{EP}_2}/\mathscr{F}_{\mathrm{Lorentz}} = \mathbf{6.80}\times$
Mode $\hat{a}_1$
$\bar{\Gamma} + \gamma_{\mathrm{PT}}$
$J = \gamma_{\mathrm{PT}}$
(EP$_2$ Locus)
Mode $\hat{a}_2$
$\bar{\Gamma} - \gamma_{\mathrm{PT}}$
$\sqrt{2\kappa_e}$
Pump $\hat{a}_0$
Nonlinear Comb Bus ($\chi^{(3)}$ Kerr / QCL / JTWPA)
$\hat{a}_+, \hat{a}_-$
$\hat{I}_+$
$\hat{I}_-$
$S_-(\Omega) = -6.4$dB
(b) Loaded EP$_2$ Coalescence & Hurwitz Boundary
Re($\lambda_\pm$)/J (Damping)
Im($\lambda_\pm$)/J (Splitting)
EP$_2$ ($J = \gamma_{\mathrm{PT}}$)
Hurwitz Limit
Eigenvalues $\lambda_\pm/J$
Normalized $\mathscr{P}\mathscr{T}$ Contrast γ_{PT}/J
1
0
−1
0
0.5
1
1.5